Advanced Assistive Brain Controlled WheelChair

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I. Abstract

In order to increase mobility for those with severe physical limitations, particularly paralysis, this research presents a unique brain-controlled wheelchair. To maneuver the wheelchair, the technology uses an optical eye-tracking device to track and translate pupil and iris movements into direction commands. A microprocessor then processes these commands. For individuals with limited mobility, it offers a practical and secure assistive solution that improves their freedom and quality of life. Multiple input and processing modules are integrated into the suggested system to enable smooth and simple wheelchair control. It improves usability and safety by incorporating obstacle avoidance and health monitoring. The prototype system was tested under controlled conditions with users performing predefined tasks, and the system has a brain-control interface, IR sensor-based eye tracking, voice commands, and an ADXL accelerometer-based sensor for easy control. The inclusion of health monitoring enhances usability and safety.

Brain-Computer Interface, Eye Tracking, EEG, Assistive Technology, Obstacle Avoidance, Health Monitoring.

Identify applicable funding agency here. If none, delete this.

II. Introduction

This composition describes a wheelchair technology that uses eye movements to enable people with physical limitations move about. The system tracks and converts pupil and iris movements into direction commands using an optic eye-shadowing device to control the wheelchair. These directives are latterly reused by a microprocessor. An accelerometer-controlled joystick is an fresh control medium for easier operation. It provides a useful and safe assistive result that enhances the quality of life for those with defined mobility. It provides a useful and safe assistive result that enhances the quality of life for those with defined mobility. The exertion of our neurons, which are connected to a variety of mortal functions including as studies, feelings, vision, and more, is reflected in EG. Because of its nature, EEG changes its point values in response to a variety of factors, including optical vision. The frequence ranges that supplement to different frequentness and define the EEG measures are called delta, theta, nascence, beta, and gamma, independently. Although a lot of work has been done to make life better for those who are paralyzed or quadriplegic, utmost of the most recent advancements are control bias that use eye blinks. still, eye blink can not be an applicable instruction since it's hard to control. By using a double eye blink and a variety of

attention situations that are both sensible by EEG, a brain-controlled wheelchair is developed. During the original phase, double eye blinks were employed as on or off commands and four attention situations were used as four- direction directives. ewline still, because they need expansive, long-term training to sustain the attention position at an horizonless range, these similar systems are less useful in real- world operations. Another study used an endless number of eye blinks as a sense and cognitive position to make a brain-computer interface that operated wirelessly and was connived using RS232C(3). In an trouble to develop a more advanced EEG- grounded wheelchair, wireless technology was used. An eye blinking signal system grounded on attention and revision was tried to be constructed in the study. In certain cases, stable- state visually elicited capabilities(SSVEP) have shown promising results(4). also, several exploration grounded on eye blink patterns are demonstrating veritably encouraging results. Eye blinks were detected using a Neurosky mindwave headset, which also counted them and transferred the data over Bluetooth to an Arduino UNO, which in turn drove a DC motor and, eventually, a wheelchair. The EEG signal was recovered and transferred to the Raspberry Pi microcontroller via the MindWave mobile operation in(6). Through a motor motorist, the Arduino Mega controls the DC motor using the joystick and the affair of the jeer Pi. They also made a virtual chart by assaying popular wheelchair routes. A many studies have formerly proposed that optical blindness might affect from simply gaping at a model(7).(8) presents the bracket of lights with varying frequentness using Steady-State Visual elicited Implicit(SSVEP) grounded exploration in combination with artificial neural networks and support vector machines. Multiclass support vector in this case. The machine outperformed the ANN when input characteristics were attained via the Fast Fourier Transform. previous exploration on wheelchair control has substantially reckoned on eye blink, SSVEP, or the Internet of effects(IoT). nonetheless, there are not numerous exploration on EEG- grounded wheelchair control at the moment. There's now exploration being done on the use of brain- computer interfaces, or BCIs. multitudinous problems with the present systems include dwindling delicacy, complex neural function under different conditions, and a lack of attention to internal conditioning. In order to manage a wheelchair, a former study used EEG data. The study set up that the beta band had the most effect while the nascence band had the least using artificial neural networks(1). This work yielded excellent issues although it only used one classifier(ANN without hyperparameter tweaking) for the bracket system and one performance index(mean square error) for system assessment(1). A balance between the two criteria is necessary since the rear tendency disregards further negative circumstances, while advanced perceptivity paired with reduced particularity lead to further wrong conclusions. Still, there are not numerous exploration on EEG- grounded wheelchair control at the moment. There's now exploration being done on the use of brain- computer relations, or BCIs. multitudinous problems with the present systems include dwindling delicacy, complex neural function under different conditions, and a lack of attention to internal conditioning. Because the EEG signal is sensitive to color cues, changes in it promise a better discovery. In order to manage a wheelchair, a former study used EEG data. It uprooted time- frequence sphere features from the signals and applied them to a color encouragement pattern. The study set up that the beta EEG band had the most effect while the nascence band had the least using artificial neural networks(1). This work yielded excellent issues indeed though it only used one classifier(ANN without hyperparameter tweaking) for the bracket system and one performance index(mean square error) for system assessment(1). still, the perceptivity and particularity criteria are the most important factors of an EEG- grounded control system.

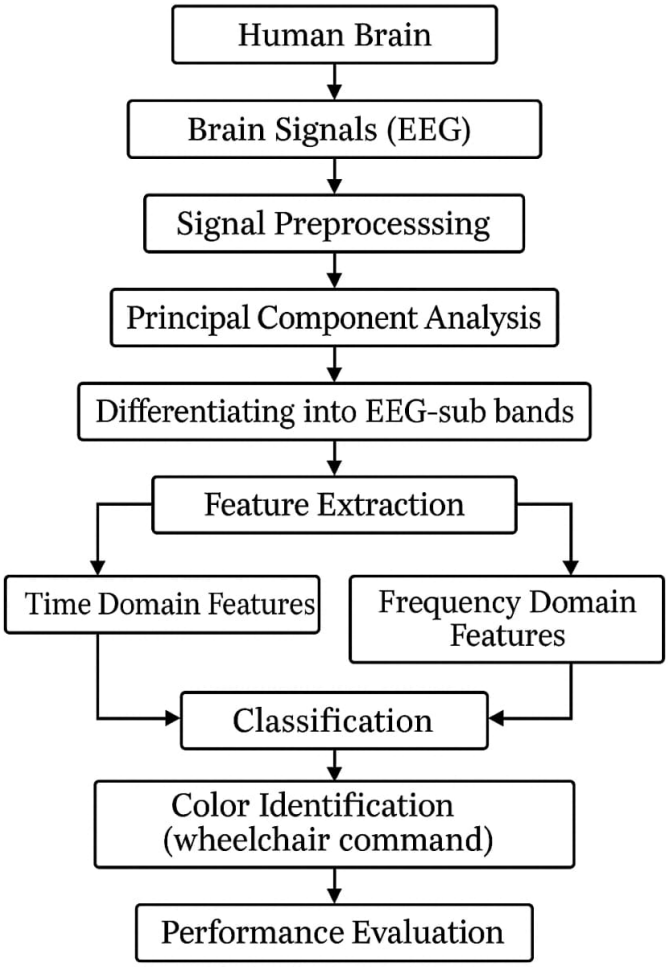

Fig. 1. brain control block diagram

III. Related Work

Research on brain-controlled wheelchairs, which use brain-wave signals to enable movement, has been ongoing. Recent

developments in brain-computer interfaces (BCIs) and electroencephalography (EEG) have greatly increased the accuracy and responsiveness of these systems. For improved control, current systems have investigated speech recognition and vision-based tracking. Nonetheless, there are still difficulties in guaranteeing smooth and secure functioning, especially in practical settings.

IV. System Design and Methodolgy

The act of input and processing modules are integrated into the suggested system to enable smooth and simple wheelchair control. This study was carried out at Universitas Padjajaran in cooperation with the Indonesian Institute of Sciences' Laboratorium of Cogno-Technology. The experiment involved 17 male, healthy volunteers who were between the ages of 17 and 22 (Turnip et al. 2019).The study's selection criterion is that participants must be almost the same time, assuming they have a comparable metabolic system that breaks down peptides and delivers them to the brain in about the same length of time. Both before and after consuming soy peptide, you are required to close your eyes and allow yourself to relax. Five minutes are recorded for each condition. Each research participant provided written informed permission and was compensated for their travel expenses. The quality of

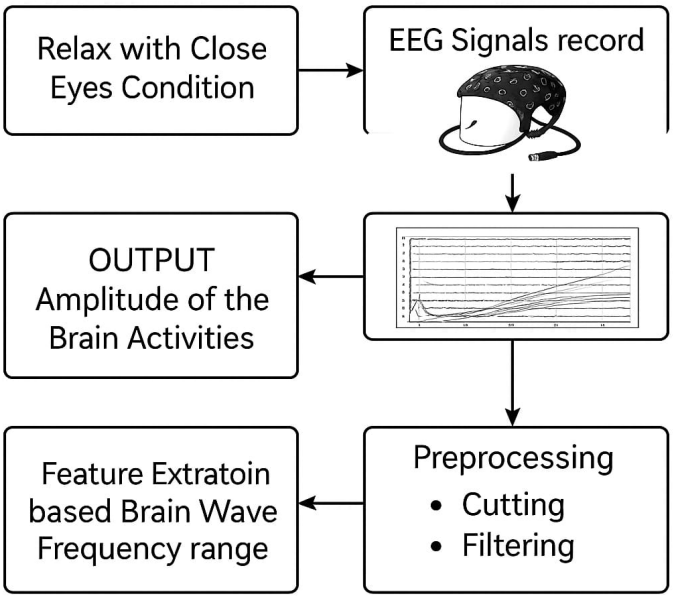

Fig. 2. Experiment and processing scheme

the recorded brain signal data in the EEG system trial is greatly told by the electrode placement pattern.However, party movement may beget vestiges or disturbances that onus the recorded data, If the electrodes are unstable or not adequately in touch with the crown. An electrogel is fitted into each electrode to increase conductivity., to more guarantee that the electrodes make contact with the crown's face. According to exploration, only 19 of the 31 channels in the EEG system used in the trial — Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, and O2 — were employed(Turnip et al. 2019). During the electrogel injection procedure, the conductivity position of the electrodes with the crown may be incontinently seen on a computer screen. Good conductivity will give a green shade in WinEEG operations(brighter colors indicate advanced conductivity and vice versa). Electro gel is added or the electrodes are dislocated slightly to alter their position if the dark shade persists after installation.In addition to conductivity, each party's head size and shape can have an impact on the quality of the EEG data that's recorded; this can be eased by ch oosing the right electro- cap size.

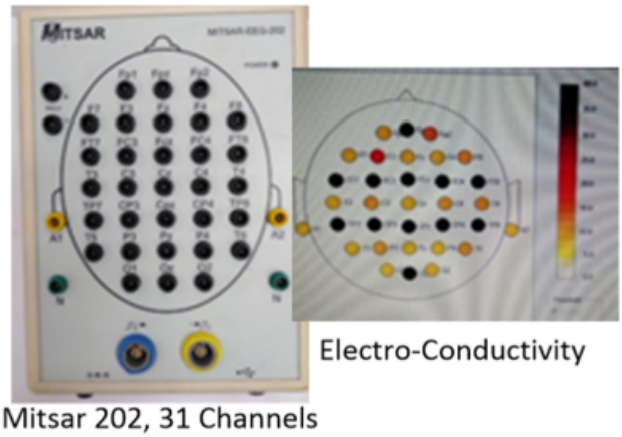

Fig. 3. EEG instrument for experiment

The recording device's specs might also have an influence on the responsibility of the data captured for operation. Electrode placement while recording is typically harmonious with the current system, specially the 10- 20 system seen in Figure 1. Each of the 19 channels named was acclimatized to four distinct brain regions anterior(F), parietal(P), occipital(O), and temporal(T). The left half of the brain is represented by odd number canons, whereas the right side is represented by indeed figures, as seen in Figure 1. Figure 2 depicts the experimental setup for probing how soy peptides affect the actors' internal health, which, among other effects, has a substantial impact on the recorded EEG signal. exertion, terrain, failing organs, external influences, and the subject's age. The frequence range of the EEG recorded data is used to identify five swells delta, theta, nascence, beta, and gamma. The 2D brain mapping pattern is also employed to interpret and estimate the EEG signal recording findings. Using two-dimensional brain mapping- reused soybean data, the tinge of the brain swells according to frequence(delta = dark blue, theta = light blue, nascence = green, gamma = maroon red, beta1 = unheroic, and beta2 = red) and normal. To assess the difference, the average power diapason of each subject's unrestricted- eye relaxation ahead and after consuming soy

peptide was calculated. The power diapason refers to the signal's intensity, or energy, at each frequence. The four separate colors were labeled for four distinct color findings, with the dependent variable enciphered from 0 to 3. The Google Colab platform employed Python 3.6.9 to produce four different bracket models K- nearest neighbors(KNN), support vector machines(SVM), artificial neural networks(ANN), and arbitrary timber classifiers(RFCs). Their performance was employed to determine the optimal classifier and meter

A. System Architechture

• **Brain-Control Interface (BCI):** Utilizes an EEG sensor (NeuroSky) to interpret user intent and convert brainwave patterns into directional commands.

• **Eye-Tracking Mechanism:** Uses infrared (IR) proximity sensors to detect iris movement and translate it into movement instructions.

• **Voice Control Module:** Employs a speech recognition system to provide alternative control for users who can vocalize commands.

• **Health Monitoring System:** Includes sensors for measuring SpO2 levels, heart rate, and body temperature to ensure real-time health tracking.

• **Obstacle Avoidance:** Uses ultrasonic sensors to detect obstacles and prevent collisions.

• **GPS and GSM Tracking:** Provides real-time location tracking and emergency alert functionalities.

• **Accelerometer-based Joystick:** An ADXL sensor is included to allow additional movement control via hand gestures.

1] ***Beta Brainwave***(13–30Hz):Out of all the waves, these have the highest frequency. They are irregular waves that form at the brain's frontal lobes. When attention is focused on cognitive and other tasks, they typically take over the typical waking states of consciousness. When someone is focused, attentive, alert, and involved in making decisions or solving problems, these waves are generated. Beta waves have also been connected to anxiety and depression because they can result in "rut-like" thought patterns.

2] ***Alpha Brainwaves*** (8–10Hz):Alpha waves are quite regular and synchronous. These waves are generated by the parietal or occipital lobes. They are created when the eyes are closed, during yoga and meditation, when the mind is concentrated, and when memory is improved. These waves are released throughout the creative process of a painter or musician. These waves help us remember things better and are created when we are relaxed. Alpha waves are often referred to as Berger waves since Hans Berger was only able to record alpha waves.

3] ***Theta Brainwaves***(4–7Hz):These waves are produced in the hippocampus area of the brain. When the mind is unhappy, these waves are created. REM sleep is when theta waves are produced. When the mind is not as awake as it is during a normal waking state, theta waves are produced just before bed and immediately after waking up. These waves are also produced during deep concentration. Recalling long-forgotten memories causes the brain to release theta waves.

3] ***Delta Brainwaves*** (2–4Hz): These waves have a number of therapeutic benefits and are produced during deep sleep. They restore the disruptions. When we wake up from a deep sleep, we are usually energized, which is why the brain produces delta waves when we are depressed or feeling anxious.

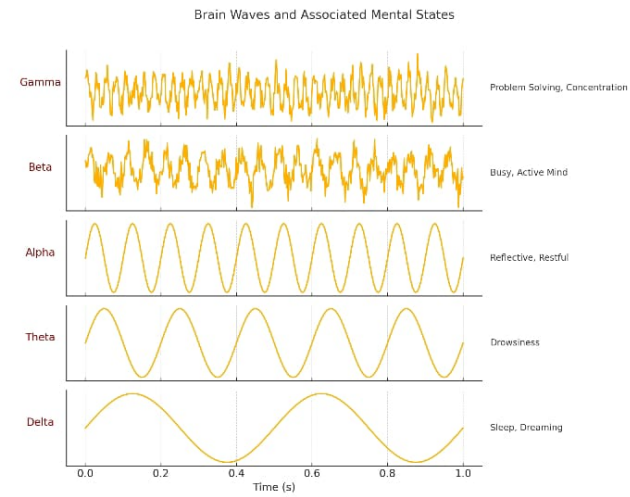

Fig. 4. brain signal and different signal parameters

Classification and Recording Location of Different Brainwaves

Rhythm	Frequency (Hz)	Amplitude (μV)	Recording & Location
Beta (β waves)	13–30	20	Parietal-frontal region main
Alpha (α waves)	8–10	50–100	Occipital region (more intense), thalamo-cortical
Theta (θ waves)	4–7	Above 50	Parietal-temporal in children Emotional stress in adults Degenerative brain states
Delta (δ waves)	2–4	Above 50	Deep sleep, infant, serious organic brain disease

Fig. 5. Tabular format brain signal

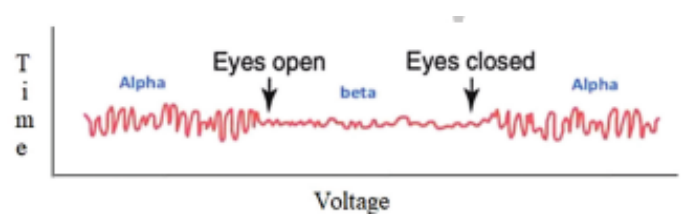

Fig. 6. Alpha block phenomena

B. Control Mechnism

The wheelchair's movement is controlled through a hierarchical decision-making process, wherein:

- EEG signals are prioritized for movement control.
- Eye-tracking inputs are processed as secondary directional commands.

- Voice commands provide additional functionality for users who can speak.
- The accelerometer sensor allows users with partial motor function to control movement using hand inclinations.

1) Results and discussions: :

The artifacts were filtered and the signals were cut, after recording the EEG data as part of an experiment. The purpose of this was to analyze changes in brain signal amplitudes (bandpass filter will be used; 0.5–70 Hz). The finding was these signals were difficult to portray and were quite severely buried in some artifacts. The raw data cut was completed throughout the extraction of features and guided by the amount of information from the records and the stated purposes of the study in detail. The feature that was collected is the mean amplitude that was extracted from the cut data results. Table 1-3 present additional detail of the extracted amplitude. In Tables 1 and 2 (subject 1), it presents the change in mean brainwave amplitude across each channel during two recording sessions, before and after eating estimated soy peptide, for 1 hour. The experiment reported in this article is in relation to relaxed recording and a blindfolded recording conditions. Because the participant was blindfolded, this study did not elaborate on delta and gamma waves, principally based on the assumption that most of the events that occur on both wavelengths are predominantly noise or noise artifact disruptions. The rest of the extracted amplitude data is found in Tables 1-3. The data that indicates mean brainwave amplitude values of each channel during the blinding recording stage one hour before and after eating the soy peptides are shown in Tables 1 and 2 (subject 1). The experiment used blinded and relaxed recording conditions. As the person was blinded, the research did not delve into depth on delta and gamma waves. This was based on the idea that noise or artifact interference is the primary cause of most activities occurring on both wavelengths. on a bipolar montage pattern in which the sensors seem to be connected to each other from the front to the back with left and right in a straight line. The intention of the study was to follow fluctuations in brain wave activity by frequency, maximal amplitude, reactivity, or highest activity by location. The alpha wave, occurring between 8 and 13 Hz, was the most recognizable among the other brain waves by frequency. It was also thought that blood flow to the brain was directly associated with the central frequency of alpha waves (the amount and rate of blood flow can vary up to 2 Hz between subjects). The changes in amplitude, the regression and variance of frequency for each channel, and wave can be observed in Tables 1 and 2. The data in tables 1 and 2 confirm that each central frequency in the bipolar derivation, except FP1-F3 and Fz-Cz, is elevated. Eye movements and mental activity are also related to these reactive or responsive number. Nevertheless, it was strongly suspected that the increases were due to the consumption of soy peptides since the participants were relaxed and had their eyes closed. The alpha rhythm changes are predominantly impacted by the state of eye movement. Aside from eye movement, the alpha rhythm is significantly impacted as a personal rhythm by concentration. Typically, where concentration increases, the alpha rhythm diminishes. While this suits the experimental situation of the study, alpha dominant rhythm is typically achieved when the subject is awake, relaxed, and has their eyes closed. The increased protein intake from soy peptides is expected to enhance brain activity, as evidenced by a shift in average middle frequency in alpha waves, which seems to represent improved blood flow through the brain. Table 3 shows the average brain wave amplitude remaining across three sample times after consuming soy peptides: before consumption, five minutes after consumption, and one hour after consumption. The trend data shows that, with the exception of delta waves, average peak amplitude for most brain wave activity tends to be higher after consumption. These observations match previous beta observations with delta waves generally being most affected by artifacts because of eye movement or other environmental and bodily distractions. Over 70 percent of the subjects in the experiment individually demonstrated an increase in average maximum amplitude. Alpha waves were statistically significant with a maximum amplitude greater than being just over 33 μV, significantly larger than the other waves, supporting that alpha waves are synonymous with brain activity. Excluding Delta waves (Table 1), the increases in maximum amplitude of Theta, Alpha, Beta 1, Beta 2, and Gamma were 3.5, 5.7, 9.8, and 24.1. The larger increases in amplitude, especially in regard to alpha waves, were more pronounced than the increases without subjects 2 and 5, who both show uniquely low amplitude data one hour after treatment consumption that can be explained due to attention issues or lack of compliance with the experimental controls.

Table 1: Pre: Amplitude of each channel of subject 1

Channels	Delta		Theta		Alpha		Beta 1		Beta 2		Gamma	
	%	Hz	%	Hz	%	Hz	%	Hz	%	Hz	%	Hz
Fp2-F8	36.54	1.46	10.12	4.15	8.77	7.57	2.43	16.36	3.77	24.17	2.93	30.27
F8-T4	24.46	1.46	8.36	3.91	17.02	10.99	4.6	17.82	7.79	22.95	6.49	30.52
T4-T6	16.12	1.71	6.25	4.39	40.84	10.99	4.15	17.58	6.32	22.95	3.81	32.23
T6-O2	15.83	1.46	9.5	4.15	47.2	10.99	3.57	13.92	4.78	21.73	2.23	30.76
Fp1-F7	30	1.46	9.64	4.15	10.13	7.32	2.66	14.16	3.19	20.02	2.09	30.03
F7-T3	22.32	1.46	7.4	4.15	17.86	10.99	3.66	19.29	5.94	23.68	4.48	29.79
T3-T5	10.39	1.46	4.71	4.15	44.21	10.99	4.66	19.78	9.63	22.46	6.95	38.09
T5-O1	13.41	1.71	7.89	4.15	54.39	10.99	3.61	13.92	5.74	29.79	2.86	29.79
Fp2-F4	42.27	1.46	9.5	4.15	7.59	7.57	2.19	17.58	2.72	20.02	1.6	30.03
F4-C4	28.07	1.46	12.78	3.91	22.26	10.99	3.72	13.92	4.29	21.73	1.98	30.27
C4-P4	19.16	1.46	7.69	3.91	43.78	10.99	3.31	13.92	3.89	22.22	1.34	30.03
P4-O2	21.02	1.46	9.27	4.15	38.79	10.99	3.54	14.4	5.43	21.73	1.77	30.76
Fp1-F3	36.22	2.2	10.51	3.91	9.32	12.45	2.67	14.16	3.24	20.02	1.83	29.79
F3-C3	26.17	1.46	11.09	3.91	29.29	12.45	3.45	13.92	4.51	21.24	1.62	30.03
C3-P3	15.47	1.46	6.98	3.91	53.94	10.99	3.13	13.92	5.01	21.73	1.4	29.79
P3-O1	14.06	1.46	7.83	5.13	54.51	10.99	3.77	13.92	6.1	21.73	1.77	29.79
Fz-Cz	24.71	1.46	12.42	5.62	28.5	12.45	3.81	13.92	4.41	23.44	1.9	30.27
Cz-Pz	24.61	1.46	9.5	3.91	31.75	10.99	3.19	13.92	3.41	21.24	1.32	31.01
Average	**23.38**	1.53	**8.97**	4.21	**31.12**	10.65	**3.45**	15.36	**5.01**	22.38	**2.69**	30.74

Table 3 provides a summary comparison of mean brain wave amplitudes among participants at three intervals: before, five minutes after, and one hour after the consumption of soy peptide. The data show a mean peak amplitude for most brain waves, other than delta waves, increases both pre- and post-consumption of soy peptide. As mentioned, delta waves are normally affected by eye artifacts, movement, and environmental stress. More than 70 subjects displayed in isolation, a mean peak amplitude increase. Of the waves studied, mean alpha waves with a peak amplitude of greater than 33 uV are larger than any other type of wave. It can be argued that alpha waves represent the most significant neural activity, aside from delta waves, theta, alpha, beta 1, beta 2, and gamma waves showed increases in amplitude of 3.5, 5.7, 9.8 and 24.1 respectively. Excluding subjects 2 and 5, the increases in amplitude were l, especially mean alpha waves. In these two subjects, we saw unusually low means for each measure at the one hour post-consumption interval, presumable because they were not vigilant or did not follow our experiment's expectations. Alpha showed the largest amplitudes, while beta 1 (under closed eye condition) showed the smallest. Figure 1 depicts a digital EEG analysis strategy called quantitative EEG (QEEG) otherwise referred to as brain mapping, which is used by Kececi and.

Table 3: Pre: Amplitude of each channel of each brain wave

Subjects	Before						After 5 min						After 1 h					
	δ	θ	α	β1	β2	γ	δ	θ	α	β1	β2	γ	δ	θ	α	β1	β2	γ
S1	23.4	9.0	31.1	3.5	5.0	2.7	16.2	7.4	31.0	9.9	8.2	6.0	21.8	9.6	34.1	3.9	5.2	2.9
S2	23.5	8.1	14.7	3.3	4.4	2.4	16.4	7.1	19.9	5.5	9.9	9.4	17.8	4.9	11.8	2.7	4.2	2.8
S3	18.2	7.1	33.7	3.7	4.4	1.8	20.3	6.0	27.0	2.8	3.2	0.6	17.1	8.3	32.8	4.7	5.5	2.7
S4	17.5	12.9	37.1	4.7	3.8	2.1	17.8	11.4	48.1	4.1	3.1	1.6	16.5	12.0	45.0	3.7	2.4	1.3
S5	14.5	8.7	34.6	7.9	5.8	4.4	13.1	8.6	24.2	5.2	7.6	1.8	14.4	8.7	11.7	7.1	7.4	7.8
S6	17.4	5.9	46.3	4.2	5.5	2.2	12.3	4.4	42.3	3.2	4.7	1.9	10.2	5.6	54.8	5.6	8.5	4.0
S7	19.5	7.0	29.1	3.8	5.0	3.2	17.3	6.4	35.4	4.2	5.8	3.9	14.4	9.6	36.5	4.3	5.7	3.3
S8	13.5	7.7	44.6	4.7	6.8	3.9	15.1	7.6	39.2	7.2	7.6	4.9	13.6	8.3	45.3	6.6	6.3	3.4
S9	10.6	7.2	63.8	4.4	3.6	1.9	10.7	7.4	62.1	3.7	3.1	1.7	10.2	5.5	57.3	3.8	3.3	2.0
S10	14.7	13.7	55.8	3.3	2.5	1.4	15.5	13.6	51.6	3.4	3.2	2.0	13.6	14.4	57.2	3.3	2.4	1.2
S11	15.6	8.0	24.5	3.4	5.4	4.2	23.4	9.0	31.1	3.5	5.0	2.7	21.8	9.6	34.1	3.8	5.2	2.7
S12	16.1	6.0	42.6	5.6	5.7	3.0	16.4	7.2	42.6	6.2	6.3	3.8	15.6	9.2	29.1	4.5	4.7	2.9
S13	14.6	9.1	35.7	4.4	7.8	3.7	12.0	6.4	23.0	2.3	2.7	1.9	13.7	7.7	28.8	3.8	6.6	1.7
S14	10.6	7.1	25.7	3.4	7.1	1.7	14.0	5.4	24.0	3.3	5.7	2.9	15.7	8.9	36.0	7.1	9.3	6.1
S15	18.2	13.6	34.0	4.7	6.2	4.3	17.3	12.2	38.3	4.7	6.1	4.4	14.0	9.8	39.9	5.3	6.8	5.6
S16	14.1	5.5	22.6	3.6	3.7	2.0	10.4	4.2	3.0	2.2	2.3	1.8	8.8	9.8	45.9	4.3	8.5	7.3
S17	21.7	9.2	31.9	3.6	3.8	3.6	2.6	7.4	25.1	2.3	2.1	2.0	18.3	9.1	36.6	3.5	3.1	3.3
Average	16.7	8.6	35.8	4.3	5.1	2.9	14.7	7.7	33.4	4.3	5.1	3.2	15.1	8.9	37.5	4.6	5.6	3.6

Table 2: Pre (after 1 h): Amplitude of each channel of subject 1

Channels	Delta		Theta		Alpha		Beta 1		Beta 2		Gamma	
	%	Hz	%	Hz	%	Hz	%	Hz	%	Hz	%	Hz
Fp2-F8	32.81	1.46	8.61	3.91	7.02	7.32	2.1	13.92	3.39	24.41	2.67	32.47
F8-T4	24.32	1.46	9.77	3.91	17.8	11.47	4.36	13.92	6.88	22.46	5.52	30.76
T4-T6	18.97	1.46	6.79	3.91	39.08	11.47	3.06	14.16	4.39	22.22	2.2	31.25
T6-O2	20.03	1.46	8.73	4.88	40.05	11.47	3.53	13.92	4.18	22.46	1.87	31.01
Fp1-F7	32.2	1.46	10.27	3.91	10.94	7.32	3.28	13.92	3.76	21.48	2.38	31.49
F7-T3	24.91	1.46	10.69	3.91	19.23	11.47	4.5	19.78	7.14	19.78	4.66	30.03
T3-T5	10.16	1.46	5.16	4.15	51.81	11.47	4.48	18.31	8.21	22.46	4.7	39.06
T5-O1	9.97	1.46	7.2	3.91	61.16	11.47	3.58	13.92	5.08	22.71	2.51	30.27
Fp2-F4	36.7	1.46	11.4	3.91	9.52	7.32	3	15.63	4.06	20.51	2.65	32.47
F4-C4	25.79	1.46	14.18	5.37	27.74	11.47	5.25	13.92	5.76	20.51	2.62	29.79
C4-P4	17.35	1.46	9.05	3.91	48.72	11.47	4.12	13.92	4.75	22.46	1.76	31.01
P4-O2	15.46	1.46	9.48	3.91	49.42	11.47	4.23	13.92	6	22.46	2.09	31.01
Fp1-F3	36.8	1.46	11.27	3.91	9.35	7.32	3.11	14.65	3.73	21	2.08	29.79
F3-C3	23.41	1.46	13.34	6.1	31.57	11.96	4.61	13.92	5.73	20.51	2.42	30.27
C3-P3	14.76	1.46	7.18	3.91	54.28	11.47	3.74	13.92	5.36	23.19	1.66	30.27
P3-O1	10.81	1.46	7.41	6.35	60.02	11.47	3.76	14.16	5.29	21.73	1.73	30.27
Fz-Cz	22.97	1.46	13.16	6.1	24.38	11.96	4.06	13.92	4.84	19.78	2.68	30.03
Cz-Pz	15.24	1.46	9.54	5.62	52	11.47	4.44	13.92	5.13	20.51	1.97	31.49
Average	**21.81**	1.46	**9.62**	4.53	**34.12**	10.60	**3.85**	14.65	**5.20**	21.70	**2.68**	31.26

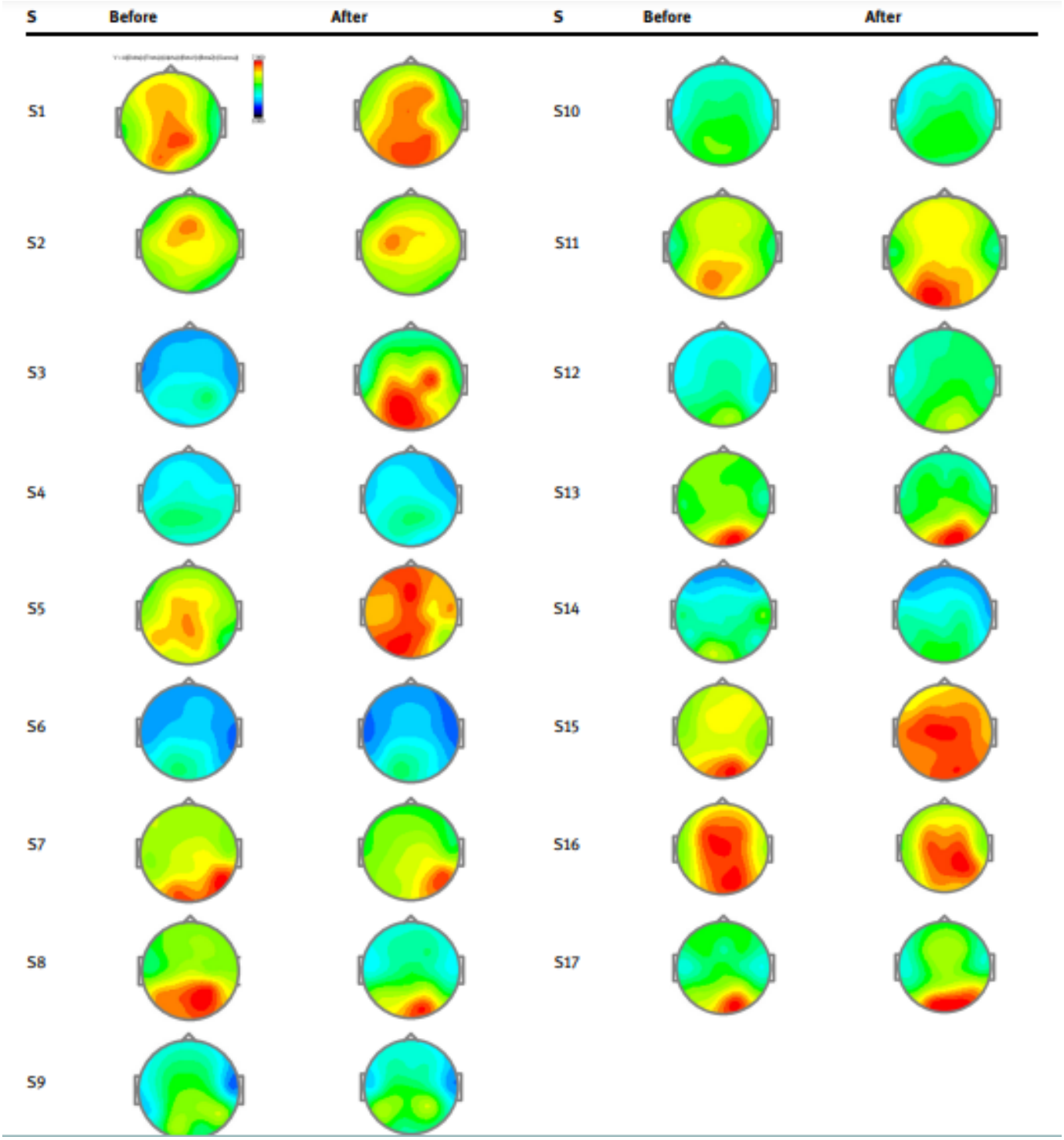

Fig. 7. brain different patterns

Consequently, QEEG metrics like absolute or relative power may be mapped to z scores and then shown in a two-dimensional brain map (Table 4). Absolute power is compared in brain maps before and after consuming soy peptide. The color-coding of the z score falls between [3, 3].

2) Results and discussions: EYE Tracking :
Eye tracking, also known as gaze estimation, is a subfield of computer vision and machine learning technologies that uses artificial intelligence (AI) technologies to follow what the user does with their eyes, also known as the gaze point, across the display. Eye tracking is the determination of where someone is looking, at any instant of time. Eye tracking can leverage existing evidence to understand behavior, including where someone is directing their attention. In some cases, eye tracking may allow tracking of a person's health.

Gaze estimation technology is based on eye position and motion tracking; that is, the location of movement of one's eyes. Eye tracking can be accomplished with a camera or other optical device (as in infrared eye trackers). Frequently, some kind of hardware needs to be added, such as eye tracking glasses. However, current gaze tracking technologies attempt to do this without additional hardware.

How Does Eye Tracking Work?
Gaze estimation works in different ways depending on the technology used.

- **Dedicated hardware** Eye movements are tracked by devices positioned beneath the monitor. To pinpoint the precise spot on the screen that the user is looking at, sophisticated systems employ a number of LEDs and cameras.
- **Wearable devices** Head-mounted sets have built-in cameras directed at the eyes.
  These cameras track pupil movement to accurately determine the gaze direction.
  Traditional eye tracking methods might rely on geometric models (e.g., PCCR—Pupil Center Corneal Reflection).

Gaze Estimation makes use of a phone, tablet, or laptop's standard camera. To determine what the user is looking at, software examines the camera image. The DAC.digital and eye square solution, which sends the camera feed to the cloud, serves as an illustration. A neural network analyzes the video in the cloud to identify the gaze point. The ability of Convolutional Neural Networks (CNNs) to identify and extract intricate patterns from eye images, including the pupil, iris, and reflections, makes them especially useful for gaze estimation. In situations where lighting, head movement, and camera angles are unpredictable, such technology is particularly helpful. In order to calibrate many gaze estimation systems, the user must look at predetermined locations on the screen. For more precise and trustworthy measurements, it enables the system to map eye position to screen coordinates.

V. Hardware Implementation

The hardware implementation consists of the following key components:

- Microcontroller: Raspberry Pi and AVR328 for signal processing.
- Motor Driver: Controls the wheelchair's motors based on processed signals.

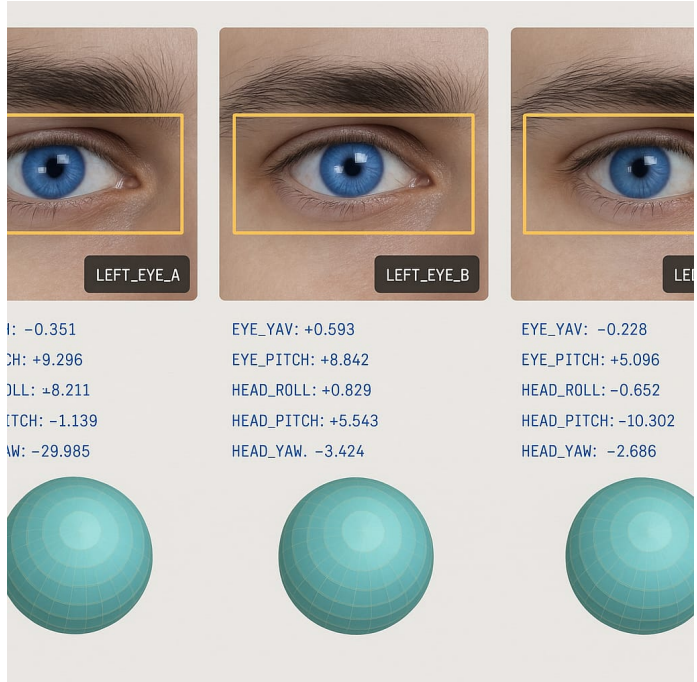

Fig. 8. EYE Different Patterns

- Sensors: Ultrasonic sensors for obstacle detection, IR sensors for eye movement tracking, and biomedical sensors for health monitoring.
- Communication Modules: GPS and GSM for location tracking and emergency alerts.

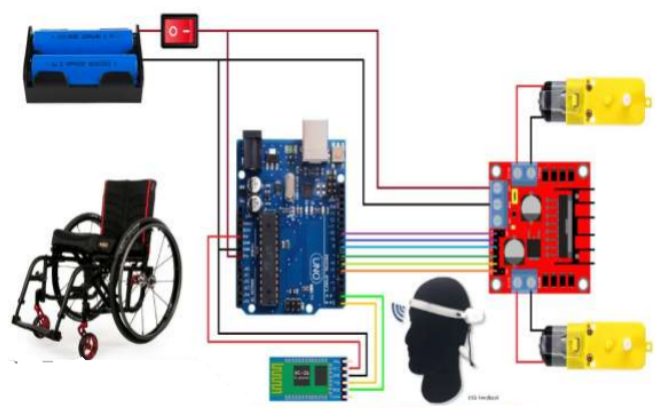

Fig. 9. brain control circuit diagram

VI. Software Implementation

Python and C are used in the system's programming, along with libraries for signal processing and machine learning-based EEG signal classification. The Google Speech-to-Text API is integrated for voice recognition, and OpenCV is used for eye tracking. Real-time decision-making and low-latency data processing are guaranteed by the system architecture.

VII. Experimental Results and Performance Evaluation

The prototype system was tested under controlled conditions with users performing predefined tasks. Key performance metrics include:

- EEG Accuracy: 85% command recognition rate.
- Eye-Tracking Precision: 92% successful detection of iris movement.
- Obstacle Avoidance Efficiency: 95% success rate in avoiding obstacles.
- Health Monitoring Accuracy: Within ±2% deviation compared to medical-grade devices.
- User Feedback: Participants reported significant improvement in independence and ease of use.

VIII. Conclusion and Future Work

A multi-modal control system that combines voice commands, eye tracking, brainwave signals, and accelerometer-based hand gestures is provided by the proposed Brain-Controlled Wheelchair. Safety and usability are improved by adding obstacle avoidance and health monitoring. Future research will focus on enhancing the EEG signal classification's resilience, incorporating AI to help with predictive movement, and carrying out extensive user trials for additional validation.

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